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ZOO502 - Animal Physiology and Behavior

Current paper solved question final term February 2020

1. How alveolar collapse is prevented? 2 marks

Prevention of Alveolar Collapse:

Alveolar collapse normally does not occur for two reasons

- Surrounding tissue prevents over-expansion of alveoli.
- Presence of pulmonary surfactants.

2. Peristalsis 2 marks

The esophagus conducts the bolus by peristaltic movement.

Peristaltic movements are the waves of contraction and relaxation in the smooth muscles of alimentary canal walls.

These movements start from oral cavity and end at the distal end of rectum.

3. Three use of crops 3 marks

In some animals, esophagus contains a sac-like expanded section, the crop, which is used to store food before digestion.

Crop is generally found in those animals that feed infrequently. It allows quantities of food to be stored for digestion at a later time

Many birds e.g. chicken.

Leeches, which feed very infrequently, with weeks or months between feeding periods.

In some animals crops are also used to ferment or digest foods for purposes other than normal digestive process. For example some parent birds prepare food in this way to be regurgitated for their nestlings e.g. pigeon's milk.

4. Write three type of blood vessels 3 marks

network of blood vessels

1) arterioles,
2) capillaries
3) venules

and a network of lymph vessels, the largest of which is the central lacteal.

Blood Vessels

Three types of blood vessels carry blood during circulation:

- • Arteries which carry blood away from the heart to organs and tissues.
- • Veins which return blood to the heart.
- • Capillaries which have thin walls formed of only a single layer of endothelial cells and permit exchange of materials between blood and tissues.

5. Mechanism of panting 3 marks

Temperature zone which activates evaporative heat-dissipating mechanisms.

Animals react to higher temperature by losing heat by active heat-dissipating mechanisms such as sweating or panting

temperatures above the upper critical temperature lead to hyperthermia. Hyperthermia occurs as the heat production and gain exceeds the rate of heat loss.

If evaporative heat loss mechanisms fail, hypothermia becomes lethal

Panting

• Panting involves use of respiratory system to lose heat by evaporative cooling. It occurs in mammals, birds and some reptiles.

• During panting, mammals inhale through the nose and exhale through the mouth, exposing the tongue and other mouth structures to encourage further water evaporation and therefore heat loss.

• It also involves increase in breathing rate and depth (hyperventilation).

• Heat is carried away in exhalant air because the dimensions of the mouth are such that exhalant air retains the heat absorbed in the lungs.

• In contrast, nasal passages and their vascularization are effective in retaining both water and body heat

6. Role of pheromones 10 marks

Pheromones:
Chemical messengers released by many animals in environment to communicate. Spread through air.
Characteristics:

a. Affect the behavior of individuals of a species
b. Effective at remarkably low concentrations
c. Have particular odors
d. Detected by olfactory receptors of receiving animal

Roles of Pheromones:

1. Recognize Members of Species:
Unicellular organisms recognize members of their own species. Many insects identify members of their colony.
2. Reproductive Roles:
Pheromones attract members of opposite sex for breeding.
Induce courtship behaviors in members of a species.
Induce spawning behavior in aquatic animals with external fertilization.

3. Territorial Behaviors:
Many male mammals mark their territories with pheromones.
It warns other males of their occupied area.
4. Insect's Social Order:
In a honeybee colony, pheromones produced by queen maintain hive's complex social order.
5. Alarm Calls:
Members of some species produce pheromones on getting injured.
It warns other members by the presence of danger in area.
6. Repel Predators:
Pheromones produce by the pray may repel predators.
Example: The skunks produce foul-smelling musk that makes skunks unpalatable or distasteful to their enemies.

7. What is kin selection give example 5 marks

Kin Selection

- Kin selection is an altruistic evolutionary strategy that favors the reproductive success of an organism's close relatives, even at a cost of organism's own survival and reproduction.
- Kin selection is an instance of inclusive fitness, which combines the number of offspring produced by an individual itself and the number of individuals it can ensure by supporting other close relatives (siblings).
- William Hamilton proposed the idea of kin (relatives) selection to explain how selection acting on related animals can affect the fitness of an individual.
- Through kin selection, a gene carried by a particular individual passes to the next generation through a related individual.
- The fitness of an individual is based on the genes it passes on as well as on those common genes that its relatives pass on.
- For kin selection to be effective, the individuals of a group must be able to identify their relatives. This is the reason that it is found in animals with advanced social structures i.e. primates and social insects.
- The altruistic kin selection behavior is most readily apparent in the act of parents sacrificing for their offspring. When parents sacrifice their own well-being to produce and aid offspring, this actually increases the fitness of the parents because it maximizes their genetic representation in the population.

8. four respiratory pigments 2 marks

A **respiratory pigment** is a molecule, such as hemoglobin in humans and other vertebrates, that increases the oxygen-carrying capacity of the blood. The four most common invertebrate

respiratory pigments are

i. hemoglobin,
ii. hemocyanin,
iii. hemerythrin

iv. chlorocruorin.

9. 4 accessory glands of alimentary canal 3 marks

Accessory Glands: Secrete digestive juices through ducts.
In mammals, the alimentary canal is associated with various accessory glands that secrete digestive juices through ducts into the gut.
The accessory glands of the mammalian digestive system are:

a. Three pairs of salivary glands
b. Gastric glands
c. Pancreas
d. Liver
e. Gallbladder

10.Aggregation with example 2 marks

Aggregation

- An aggregation is any form of gathering of animals or the process of coming together.
- There is minimal interaction between members of an aggregation.

For example, several Drosophila flies living on a piece of rotting fruit

11.Difference between cortex and medulla 3 marks

Cortex and medulla

The kidney is divided into two functional layers that contain the functional urine forming units— nephrons. The outer layer is called renal cortex while the inner layer is renal medulla.

12.Torpor with example 5 marks

Torpor

Small endotherms can not maintain their high rates of metabolic rates when they are not feeding during periods of inactivity.

During those periods, some of these animals enter a state of torpor, in which body temperature and metabolic rate decrease.

The lower basal metabolism lowers the rate of conversion of energy stores into body heat.

Thus, it is generally advantageous to allow body temperature to decrease during periods of nonfeeding and inactivity.

As the animal comes out of torpor and becomes active, it undergoes a burst of metabolic activity, through shivering or oxidation of brown-fat and its body temperature rises.

Daily torpor is practiced by many terrestrial birds. The hummingbird is a classic example, allowing its body temperature to fall from a daytime level of about 40°C to a nighttime level as low as 13°C.

Several species of small mammals also undergo torpor (e.g., shrews).

13.Role of Appendix in digestion and function 3 marks

The appendix is a finger-like, blind-ended tube connected to the cecum, from which it develops in the embryo. The cecum is a pouch-like structure of the colon, located at the junction of the small and the large intestines.

Appendix has no role in digestion or absorption but has an undefined role in immunity as it has many lymph nodes

14.Types of stomach 5 marks

Types of Stomach in Vertebrates

Based on the number of chambers, stomachs are classified as:

- Monogastric stomachs
- Digastric stomachs

Monogastric Stomachs

A Monogastric stomach consists of a single strong muscular tube or sac.

Carnivorous and omnivorous vertebrates have a Monogastric stomach.

Digastric Stomachs

Digastric stomachs are multi-chambered and are found in ruminant mammals of order Artiodactyls e.g. deer, giraffe, sheep, cattle, camel, llama etc.

These stomachs have four chambers which are separated into two divisions.

- The first division consists of fermentation chambers called rumen and reticulum.

The second division comprises the omasum and abomasum (true stomach).

15.4 steps of gas transport system 5 marks

A gas-transfer system in animals takes in oxygen from the environment takes it to the issues through blood and brings CO2 from tissues, transporting it to the gas exchange surface and removes it by diffusion.

Such a system involves four basic steps:

1. Breathing movements, which assure a continual supply of air or water to the respiratory surface (e.g., lungs or gills)
2. Diffusion of O2 and CO2 across the respiratory epithelium
3. Bulk transport of gases by the blood
4. Diffusion of O2 and CO2 across capillary walls between blood and mitochondria in tissue cells

16.Apoenzyme and function

Apoenzymes- An enzyme that requires a cofactor but does not have one bound. An apoenzyme is an inactive enzyme, activation of the enzyme occurs upon binding of an organic or inorganic cofactor. The term holoenzyme is used to refer to the apoenzyme bound to a cofactor.

Or

Apoenzymes refers to inactive form of enzyme which activates upon the binding of a cofactor.

Function;

It is responsible for reaction

17.Define cognition map 3 marks

Cognitive Maps is also called mental map

Some animals guide their activity by a cognitive map, a representation in the nervous system of the spatial relationships between objects in an animal's surroundings. Rather than relying solely on moving from landmark to landmark, animals using cognitive maps can navigate more flexibly and efficiently by relating landmark positions to one another.

It is essential in environment wehere strategic decisions are made in rapidly shifting economic environment

18.Neural plasticity and explain it with example 10 marks

Although the basic architecture of the nervous system is established during embryonic development, the behavioral patterns can modify with experience. This capacity of nervous system to change with experience is known as neuronal plasticity.

Neuronal plasticity is of premier importance for the survival of any organism.

Examples of Neural Plasticity

Learning, memory and development of motor skills and habits.

- Neural plasticity lies behind human intelligence.
- The ability of all higher animals to respond to stimuli in diverse ways, different from fixed and programmed reflexes.
- Behavioral plasticity, virtually demonstrated by all animals.

Mechanism of Neuronal Plasticity

Much of the reshaping of the nervous system occurs at synapses.

The mechanisms that underlie synaptic plasticity are currently the subject of many experiments.

Modifying Synapses—Synaptic Plasticity

Synaptic plasticity may takes place as a result of developmental events over the course of a lifetime. Synaptic connections that are established in embryos are later refined into adult patterns.

19.Objectives of ethology 3 marks

Ethology: "Scientific study of animal behavior in natural environment

Objectives of Ethology: Involves understanding of

1. Understanding the stimuli that elicit behavior.
2. Understanding the physiological mechanisms that mediate the response.
3. Understanding how an animal's experience during growth and development influence the response.
4. Understanding how the behavior aids in survival and reproduction.
5. Understanding the behavior's evolutionary history.

20.What is Blubber and its role 2 marks

Blubber is a thick layer of fat, also called adipose tissue, directly under the skin of all marine Mammals.

Role: 1. Blubber is a good insulator because it has a lower thermal conductivity.

2. Fatty tissues are metabolically very inactive and have little blood vascularization. This 3333prevents heat loss through body surface.
3. It stores energy, insulates heat and buoyancy.

21.Fahreous Lindquist effect 2 marks

Fahraeus-Lindqvist Effect:

Phenomenon of plasma skimming & resultant increased blood flow velocity is called Fahreous Lindquist effect. It reduces energy required to drive blood through microcirculation.

22.Hepatocytes nutrients transport 3 marks

Transport of Nutrients

Digestion products absorbed enter either the blood or the lymphatic circulation in the villi.

Blood Transport of Sugars and Amino Acids

- Sugars and amino acids primarily enter the capillaries of the villi.
- These capillaries are drained by venules that lead into the hepatic portal vein.
- The hepatic portal vein takes the blood from the intestine directly to the liver.
- In the liver, much of the glucose is taken up by hepatocytes and, under the influence of insulin, is converted into glycogen granules for storage. Only a controlled quantity is released into the circulation to be utilized by tissues.

23.Write note on types of stomach vertebrates 5 marks

Types of Stomach in Vertebrates

Based on the number of chambers, stomachs are classified as:

- Monogastric stomachs
- Digastric stomachs

Monogastric Stomachs

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24.Hypoxia in an aquatic environment 5 marks

Hypoxia

• oxygen deficiency in biotic envirоonment“ oxygen depletin in aqutic environment as dissolved oxygen.

Hypoxia in Aquatic Environment

• O2 levels in aquatic environment vary more frequently

• Aquatic animals are faced with frequent and rapid changes in oxygen levels. Many aquatic Animals can withstand very long periods of hypoxia

Respiratory Responses of Aquatic Animals to Hypoxia

- Oxygen level in aquatic environment varies frequently. Aquatic animals are subjected to more frequent and rapid changes in oxygen levels.
- Many aquatic animals can withstand very long periods of hypoxia. Many of these animals utilize a variety of anaerobic metabolic pathways to survive the period of reduced oxygen availability.
- Many animals adjust the respiratory and cardiovascular systems to maintain oxygen delivery during reduced oxygen availability. For example, many fishes increase gill ventilation rate in hypoxic conditions.
- Ram ventilators (e.g. tuna) swim forward with their mouth open, increase the size of gap during hypoxia to increase water flow over the gills.

25.Panting in mammals 3 marks

Panting

• Panting a method of cooling used by many mammals, birds and some reptiles involves use of respiratory system to loss heat.

• During panting, mammals inhale through the nose and exhale through the mouth, exposing the tongue and other mouth structures to encourage further water evaporation and therefore heat loss.

• It also involves increase in breathing rate and depth (hyperventilation).

• Heat is carried away in exhalant air because the dimensions of the mouth are such that exhalant air retains the heat absorbed in the lungs.

• In contrast, nasal passages and their vascularization are effective in retaining both water and body heat.

26.Parental care 5 marks

Parental Care in Mammals

• In mammals, lactating female nourishes the young and is involved in most of the caring behaviors. Males usually play no role in raising the young.

• In some mammalian species, such as lions, males protect the females and young. In such species a male or small group of males takes care of many females at once in a harem.

27.Composition of bile 2 marks

Bile:

Vertebrate liver secretes bile. Bile doesn't contain any digestive enzymes. It contains many salts Bile salts are essential for the digestion of fats

Composition of Bile:

Bile consists of water and a weakly basic mixture of cholesterol, lecithin, inorganic salts, organic salts, and bile pigments.

The bile organic salts are manufactured by the liver from cholesterol and amino acids complexed with sodium. The bile pigments are derived from biliverdin and bilirubin, which are products of the breakdown of hemoglobin coming from ruptured red blood cells.

28.How to nervous system maintain blood pressure 5 marks

Role of Nervous System in Maintaining Blood Pressure

Nervous control serves to maintain arterial pressure by:

• Adjusting resistance to blood flow in the peripheral circulation.

- controlling the number of capillaries open at any moment

The nervous control of capillary flow ensures that if arterial pressure falls, blood flow to the gut, liver, and muscles is reduced to maintain flow to the brain and heart.

29.Function and parts of colon 5 marks

Colon

- The colon is about 1.5 m long and consists of four parts: ascending colon, transverse colon, descending colon and sigmoid colon.
- Colon receives the undigested part of the food along with inorganic ions and excess water.
- Ions and water are absorbed by the colon.
- As water is absorbed, indigestible wastes become increasingly solid and are converted into feces as they move along the colon by peristalsis.
- It takes approximately 12-24 hours for material to travel the length of the colon and reach the rectum for elimination through the anus.

30. Vasodilation

vesodilation is the widening of blood vessels.

- o Decreased tissue oxygen levels or increased CO_2 levels.
- o Lactic acid or other acids generated by tissue cells.
- o Nitric oxide (NO) released from endothelial cells.
- o Rising concentrations of potassium ions or hydrogen ions in the interstitial fluid.
- o Elevated local temperature.

31.Which enzyme involved in pancreatic secretions

Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

Pancreas produces several proteases which are secreted in the pancreatic juice.

32.What is proenzyme with example

Proenzymes

Certain digestive enzymes, in particular proteolytic enzymes, are synthesized, stored, and released in an inactive molecular form known as a proenzyme or zymogen.
Packaging of the enzyme in an inactive form prevents self-digestion of releasing tissues/cells by the enzyme.
The proenzyme is activated by the removal of a portion of the molecule, either by the action of another enzyme specific for this purpose or through a rise in ambient acidity.

Examples

Pepsinogen-Pepsin
Pepsin is the gastric proteolytic enzyme.
Trypsinogen-Trypsin
Trypsinogen is secreted in pancreatic juice as a proenzyme.
Chymotrypsinogen-Chymotrypsin
Chymotrypsinogen is also secreted as proenzyme in pancreatic juice.

33.Animal orientation detail 5 marks

Animal Orientation

Orientation is the ability of an animal to determine its position in space and move predictably with respect to specific stimuli.
The best examples of animal orientation are observed in response to gravity. A human who has fell down on ground orients himself by standing up. Some animals orient themselves towards the incoming light. Other examples include acoustic and olfactory orientations.

Sensory Motor Basis of Orientation

Animal orientation is a complex process that requires the integration of sensory input and the coordination of motor output.

It includes receiving information through the sensory receptor neurons, processing and correlating it in the central nervous system and forming a response through the muscles that cause the body of animal to move.

Types of Orienting Movements

1. Kinesis
2. taxis

Kinesis

Kinesis is a change in activity or turning rate in response to a change in location.

Taxis

A taxis is an oriented movement toward (positive taxis) or away (negative taxis) from some stimulus.

34.Orthenine urea cycle 5 marks

Ornithine-Urea Cycle

• All vertebrates except most teleost fishes, synthesize urea primarily in the liver via the ornithine-urea cycle.
• In this cycle, two ammonia and one CO2 molecule are added to ornithine to form arginine.
• Arginine is cleaved by the enzyme arginase, to remove a molecule of urea and regenerate ornithine.

35.Type of alimentary system 2

Types of Alimentary Systems:

Two basic plans of alimentary systems in animals:

A. Sac-like gastro vascular cavity
B. Tubular alimentary canal or gut

36.Baroreceptors 3 marks

Baroreceptors

The baroreceptor system is a simple and rapidly acting control mechanism that contributes to the regulation of arterial blood pressure

Baroreceptors in Vertebrates

Baroreceptors are widely distributed in the arterial system of vertebrates.

• The central cardiovascular system of amphibia, reptiles and mammals has unmyelinated baroreceptors.

• These unmyelinated baroreceptors only respond to pressures above normal, initiating reflexes that reduce arterial blood pressure and thus protect the animal from damaging increases in blood pressure.

37.Three ways of survival in cellular freezing. 3 marks

Such animals employ three methods to survive freezing environments:

• Ice-nucleating Agents (INAs)
• Cryoprotectants (antifreeze substances)
• Supercooling

38. What is core temperature

The core temperatures are maintained nearly constant between 37°C and 38°C in mammals and about 40°C in birds.

39. Dose vertebrate have lungs

All vertebrate animals that live on land have lungs. When we breathe in, the muscle below the rib cage (called the diaphragm) is pulled down, and air gets sucked into the rib cage, filling the lungs. Blood cells circulating through tiny blood vessels near the lungs pick up oxygen and carry it around the body to the sites of respiration. Air is then forced out of the lungs as the diaphragm bows upwards

40. What is Rete Mirabile? Mention its function in the Tuna fish? 5 marks

Rete Mirabile

A highly evolved system of countercurrent heat exchange. Found widely in homeothermic and heterothermic vertebrates. The artery carrying warm blood toward extremity is completely encased in a circlet of veins carrying back cold blood.

Function in the Tuna fish:

Forms an extensive heat exchange surface b/w inflowing and outflowing blood. Large number of rete mirabile in tuna (fish) are used to regulate temperature of brain, muscles, and eyes. Swimbladders of some fishes (e.g. eels) are associated with rete mirabile and functioning as CO2 exchangers.

A countercurrent network arrangement of arterial and venous capillaries is referred to as a rete mirabile.
Before entering a tissue, an artery divides into a large number of small capillaries that parallel a series of venous capillaries leaving the tissue.
The arterial capillaries are surrounded by venous capillaries.

Rete Mirabile—Functions

This arrangement forms an extensive exchange surface between inflowing and outflowing blood.

Examples

Tuna (fish) have a large number of rete mirabile, which are used to regulate the temperature of the brain, muscles, and eyes.
The rete mirabile leading to the swimbladder of some fish (e.g. eels) function as a carbon dioxide countercurrent exchanger.

41.Hormones that control the pancreatic secretion

Pancreatic secretions are controlled by a variety of hormones including peptide hormones, gastrin, cholecystokinin, somatostatin and enkephalins.

The secretion of intestinal juice is regulated by several hormones, including secretin, gastric inhibitory peptide (GIP), and **gastrin**.

Gastrin secreted from the stomach lining also elicits a small flow of pancreatic juice in anticipation of the food coming in the duodenum.

42. Types of alimentary canal 2 marks

Regions of the Alimentary Canal

In general, alimentary canals can be divided on a structural and functional basis into four major divisions:

(1) **Headgut**: specialized for receiving ingested material. Includes Buccal cavity and pharynx

(2) **Foregut**: specialized for conducting, storing, and digesting ingested material. Includes esophagus and stomach. Also includes crop in some animals.

(3) **Midgut**: specialized for digesting and absorbing nutrients. Includes small intestine: duodenum, jejunum and ileum.

(4) **Hindgut**: specialized for absorbing water and defecating. Includes large intestine: colon, cecum, rectum and anus.

43.How gravity effect on arterial blood pressure 3 marks

Effect of Gravity on Arterial blood Pressure

Gravity affects the column of blood in larger arteries. So pressure in limbs and head region varies in different positions of the body.
In lying down position, the heart is at the same level as the limbs and head with respect to gravity. So, blood pressure in the arteries of head and limbs is equal.
In a standing position, the relationship between the head, heart, and limbs changes with respect to gravity.
In this position heart is below the head and about a meter above the lower limbs. So, the arterial pressure in lower limbs increases while in the head region, it decreases.

Effect of Gravity on Venous Flow

Because the vascular system is elastic, an increase in absolute pressure expands blood vessels, particularly the veins which are more compliant.
Thus, pooling of blood tends to occur in veins, in different regions of the body

44.Role of nervous system in regulating blood pressure 5 marks

Role of Nervous System in Maintaining Blood Pressure

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- Adjusting resistance to blood flow in the peripheral circulation.
- controlling the number of capillaries open at any moment

The nervous control of capillary flow ensures that if arterial pressure falls, blood flow to the gut, liver, and muscles is reduced to maintain flow to the brain and heart.

Nervous Innervations of Arterioles

• Most arterioles are innervated by sympathetic nerves, which release norepinephrine at their endings.
• Some arterioles are innervated by parasympathetic nerves, which release acetylcholine at their endings.

Sympathetic Nerve Stimulation

• Stimulation of sympathetic nerves causes release of norepinephrine.
• Binding of the norepinephrine to α-adrenoreceptors in the smooth muscle of arterioles causes vasoconstriction and therefore a decrease in diameter of the arterioles.
This decrease in diameter causes an increase in resistance to flow, thus reducing blood flow through that capillary bed.

45.Three local vasodilation

Vasodilation is the widening of blood vessels. It results from relaxation of smooth muscle cells within the vessel walls, in particular in the large veins, large arteries, and smaller arterioles. ...

- sildenafil (Viagra)
- tadalafil (Cialis)
- vardenafil (Levitra)

46.Chemoreceptors

chemoreceptors in our bodies, sensory cells or organs that interact with chemicals in our blood and in what we eat and smell. These chemoreceptors then convey messages about what they've found to our brains to get a response. These responses may tell us not eat something that's rotten or to flee from the smell of smoke. Our heartbeat and respiration rates are also controlled by chemoreceptors that detect carbon dioxide, oxygen, and pH levels in the blood.

Three types of chemoreceptors monitor changes in O2 and CO2 and pH levels in arterial blood:
• Peripheral Chemoreceptors
• Central Chemoreceptors
• Type-J Chemoreceptors

BEST OF LUCK